

SandCASTLE:

A Portable, Economical, Environmentally Friendly Habitat Printer for Low-Resource Communities

Introduction

One of the biggest problems faced by humans is a worldwide lack of affordable housing. Existing house-building methods are slow and costly, require skilled labor, and all building materials must be transported to the site. This is especially challenging for low-resource or rural communities. Alternative technologies for building houses exist, namely 3D printing houses using cement,¹⁻³ but this technology also has its downsides that can potentially be improved upon. Because the 3D printer has to extrude 100% of the house material, the gantry (Figure 1) must be large and strong, making these machines quite expensive.⁴ Additionally, the precise qualities of cement needed for this application make cement procurement similarly expensive. Cement 3D printers are also bad for the environment, as their main ingredient, Portland cement, is very harmful due to its significant carbon footprint, air pollution generated during production, and high water use.⁵ Finally, like traditional building methods, there is still a requirement to source materials elsewhere and transport to the building site.



Figure 1: Vulcan, an example of existing house 3D printing technologies from the company ICON.

In this project, I wanted to develop an alternative method for 3D printing houses that keeps the benefits of existing approaches while also addressing some of their flaws.

Project Goals

- Design and construct a 3D printing rig using lightweight and inexpensive components
- Create a program that can turn a 3D object into code that can be run on the printer
- Printer should:
 - Move in a 3D coordinate system to allow for precise glue application
 - Smooth out sand into layers
 - Dispense the glue
 - Create a 3D structure

Advantages

- Uses on-site materials (sand)
- Lighter frame for 3D structure printer
- Minimizes transportation costs (only glue)
- Ability to print overhangs (print roof, not just walls)
- Substitution of glue instead of cement decreases environmental impact

Basic operation



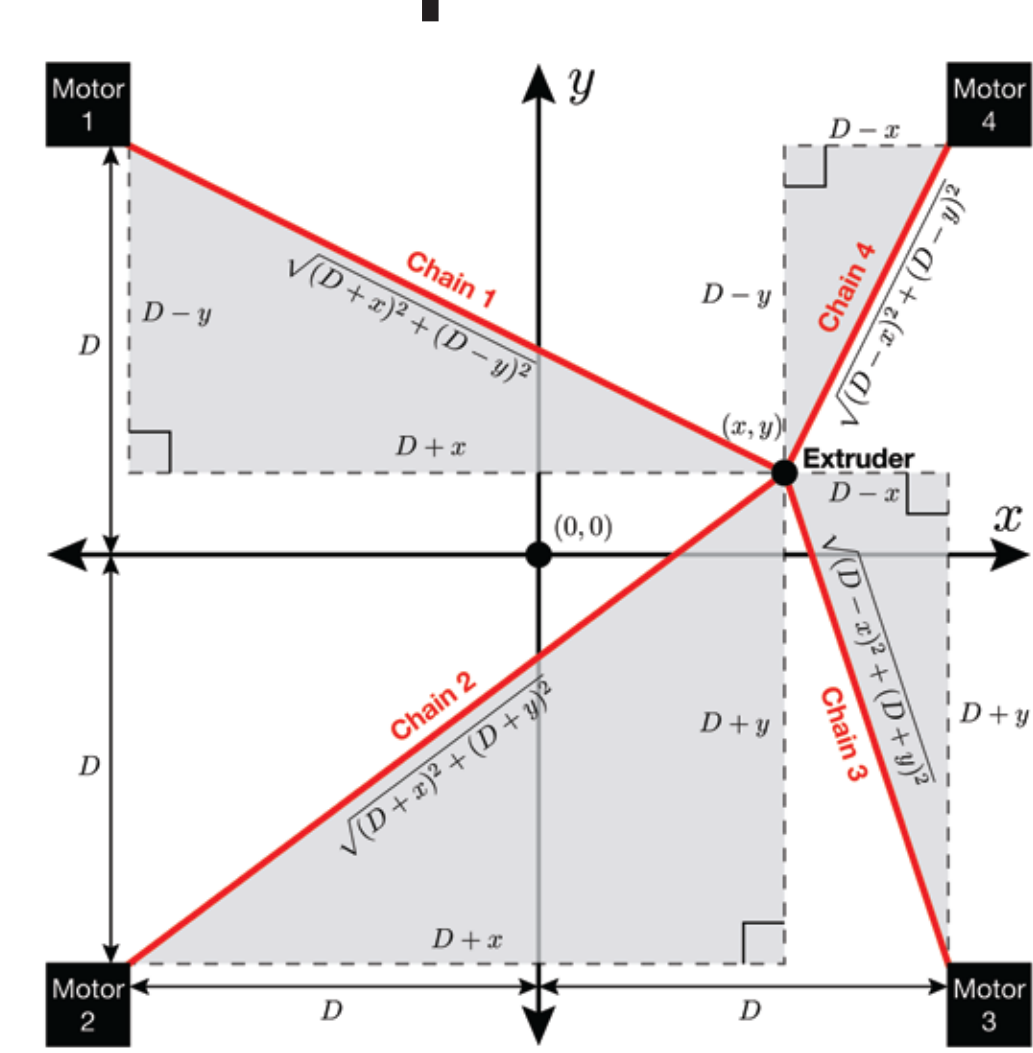
Glue Type Selection



Figure 2: Preliminary tests done to determine strength capabilities of different glue types. (a,b) Casting glue+sand bars. (c) Strength testing rig. (d) Results from strength testing.

Calculations

Top View



Side View

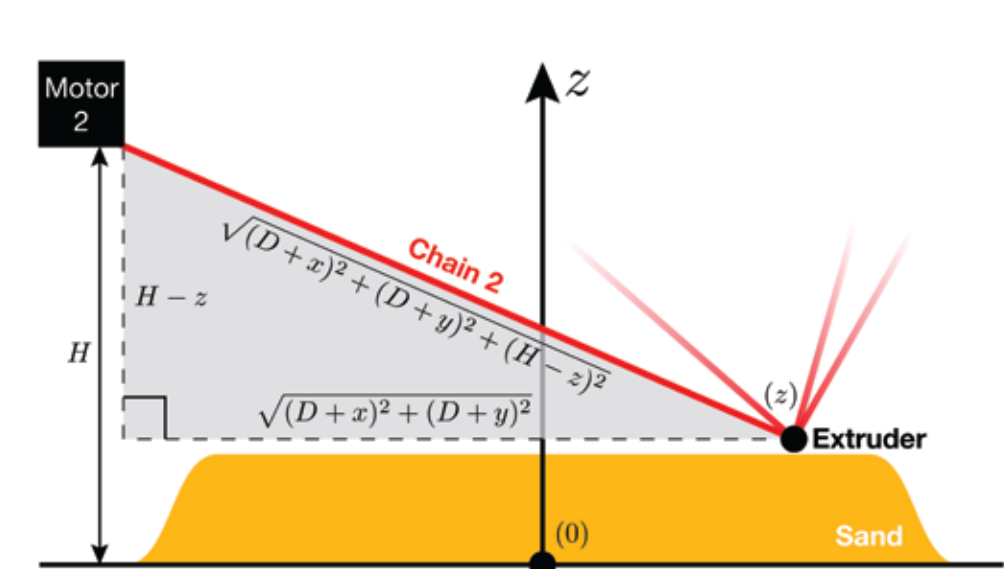


Figure 3: Top and side views of simplified calculations done to convert X,Y,Z coordinates into target lengths of chain for each stepper motor.

2D Printer Parameter Optimization

Resin Comparison

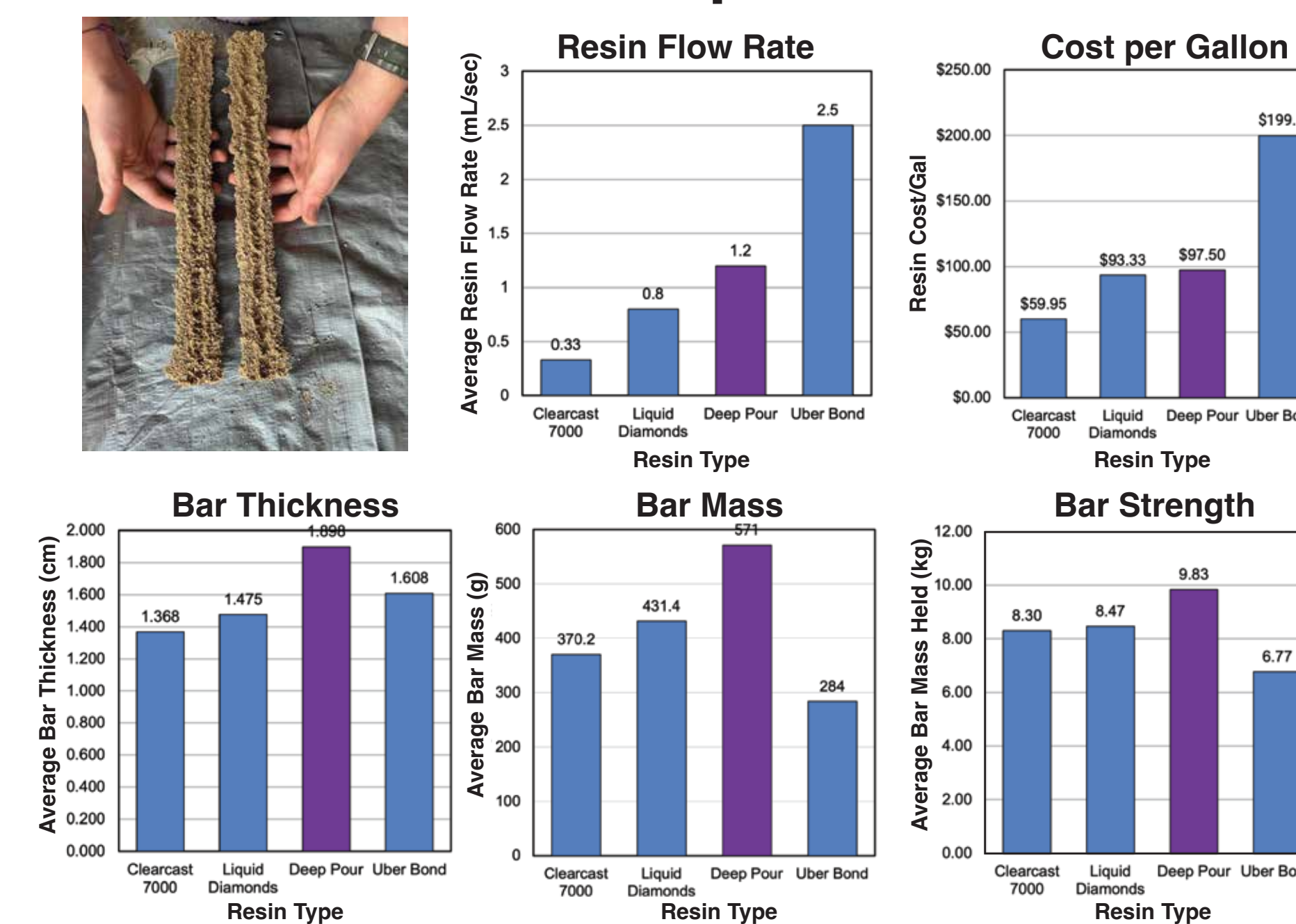


Figure 4: (Top) Comparison of flow rate and cost of different types of resin. (Bottom) Characteristics of bars made with 40 g of each resin.

Speed and Layer Thickness Optimization

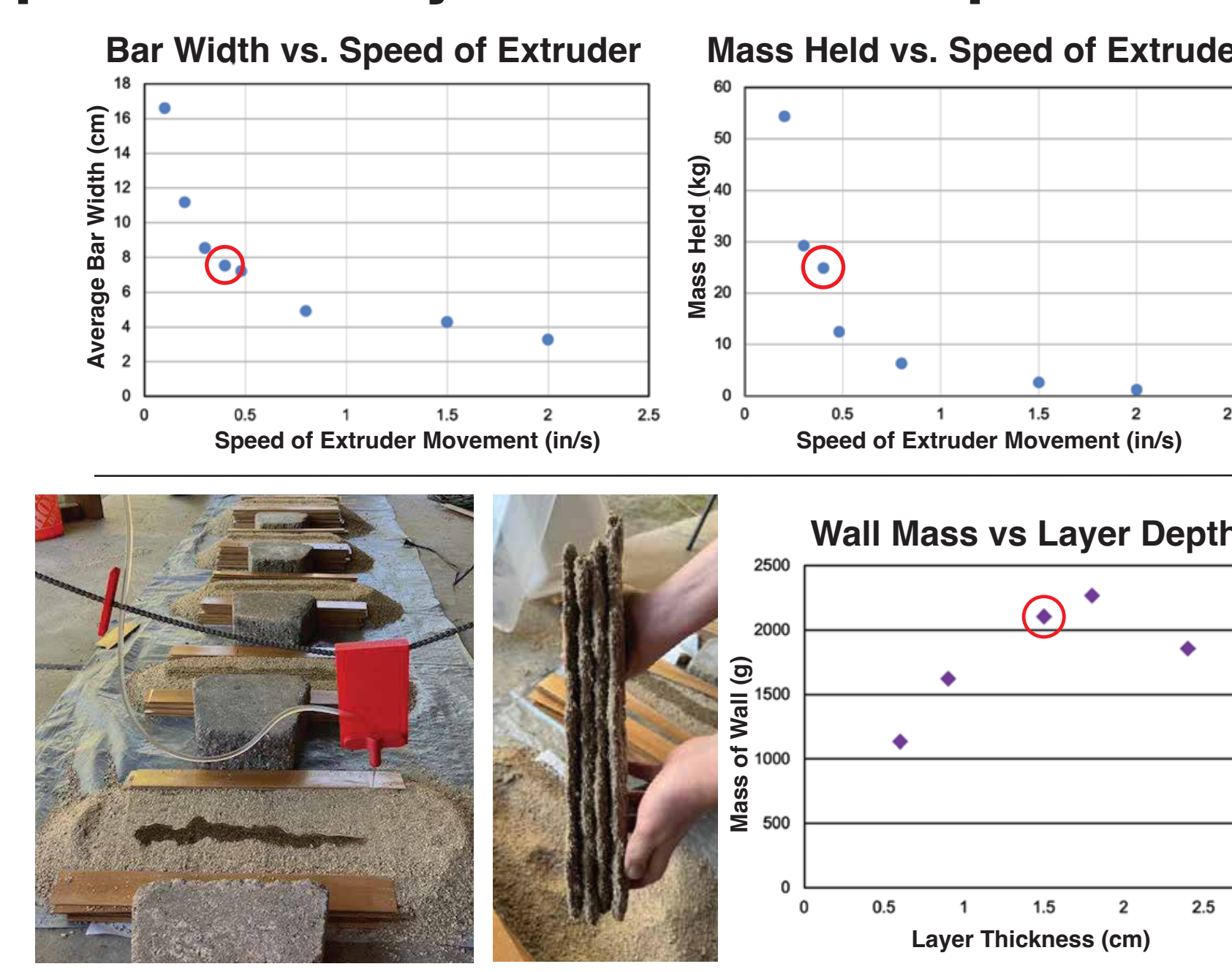


Figure 5: (Top) Optimizing extruder speed based on width and mass held by glue bars. (Bottom) Determining optimal layer depth.

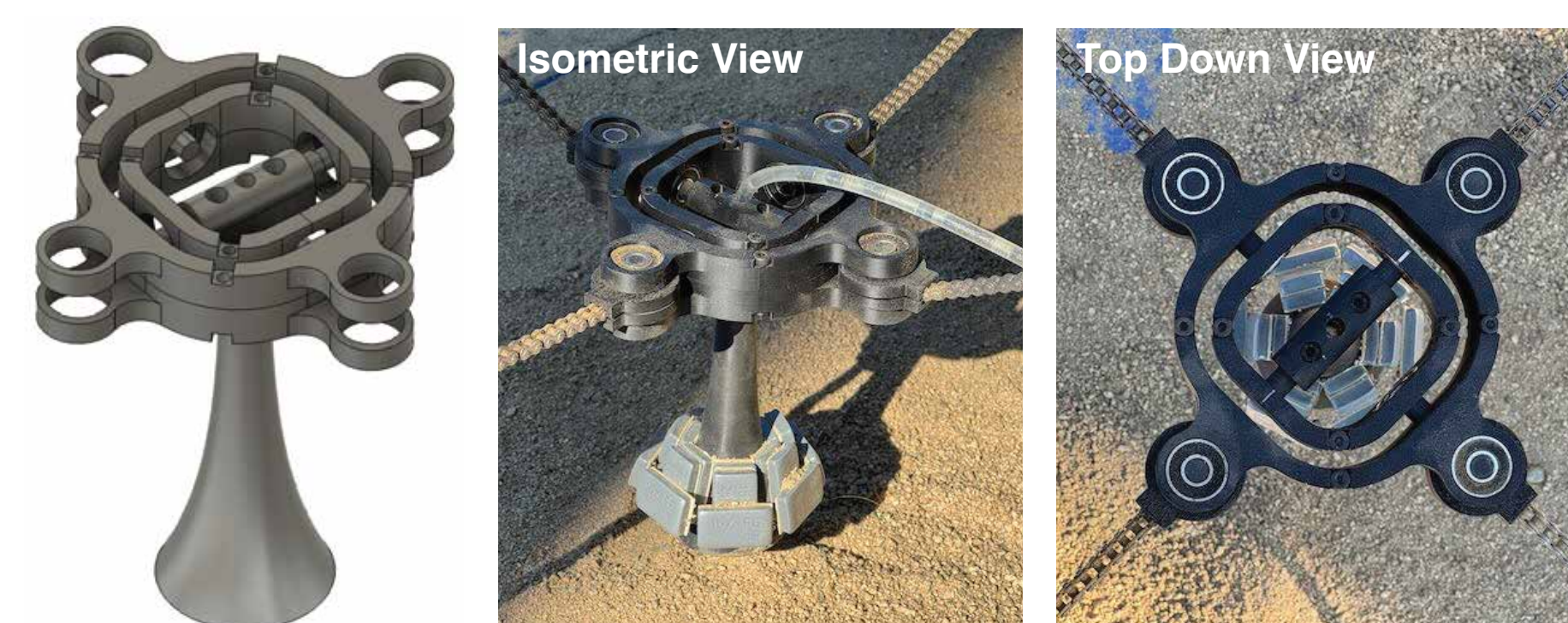
3D Printer - Rig and Components

Proof of Concept: 3D Printing Rig



Figure 6: Images of the constructed proof of concept rig, consisting of pop-up tent frame with custom 3D printed components attached on 4 legs. Chains (white arrows) from each of the 4 motor components (red circles) connect to a central extruder (white circle).

Custom Designed Pump Head



Custom Designed Stepper Motor Mounts

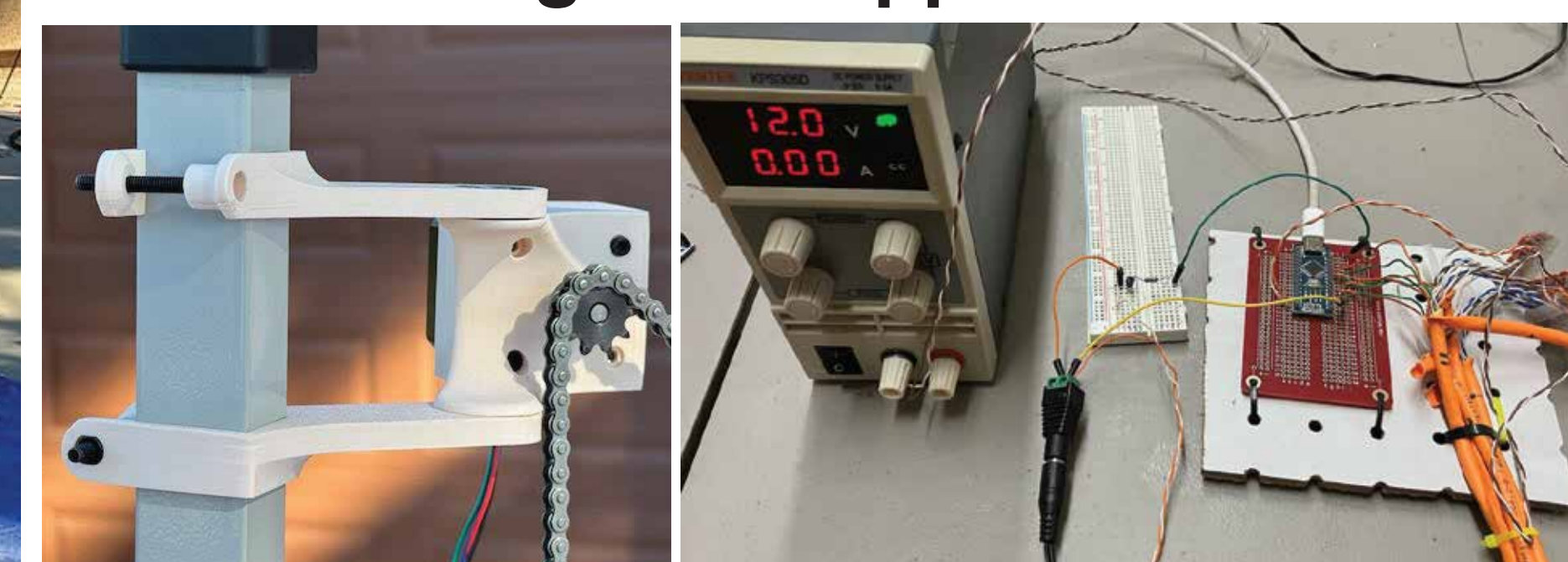


Figure 7: (Top) CAD design and images of the extruder pump head including 4 bearing connections that attach to the motor chains. (Bottom left) Stepper motor mount. (Bottom right) Power supply and Arduino control for 4 stepper motors and pump.

3D Printed Structures

Design 1: Hollow Dome



Design 2: Bench



Design 3: Scale Model House



Discussion/Conclusion

This project successfully demonstrates the principles of sand-resin 3D printing, where only 7.5% of the final house's material is extruded and the remaining 92.5% is ordinary sand from the building site, which significantly minimizes material and transportation costs. This method creates walls that can be as cheap as \$1.28/sq.ft of wall, where traditional methods can range from \$7-\$45 per sq. ft. of wall, depending upon materials.⁶

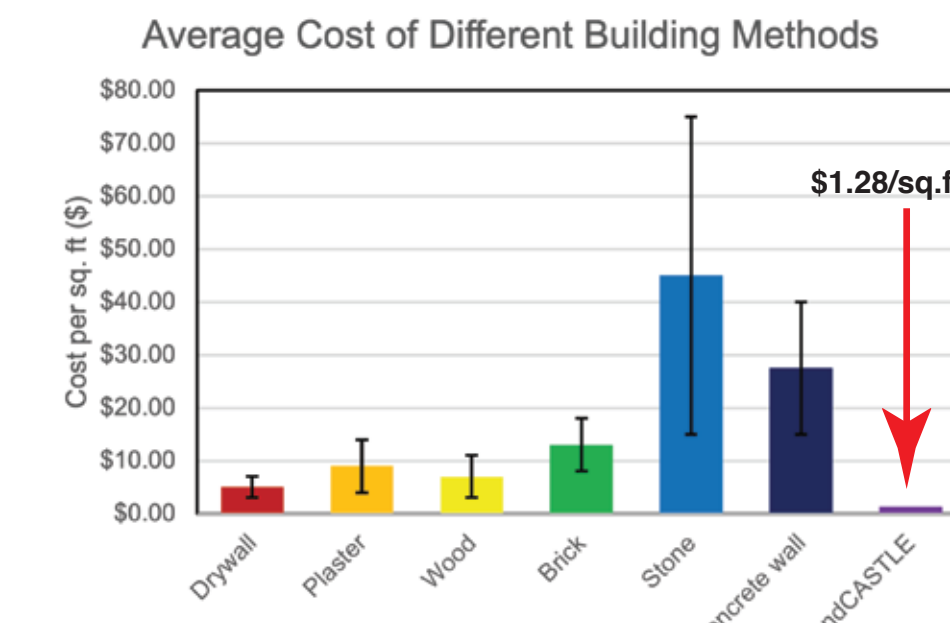


Figure 8: Cost to build walls using different building methods.

Furthermore, because a smaller percentage of the house's materials are moved by the printer, the printer design is simpler and the electricity costs are minimal. This means that the robot itself is very inexpensive, costing < \$270, while existing house-sized 3D printers cost hundreds of thousands of dollars.³ In addition, the materials themselves are more sustainable than concrete (which generates 7-9% of all human CO₂ production per year⁶).

	SandCASTLE	Cement 3D
% of Materials Transported	<7.5%	100%
Weight of Materials Transported	Very low	Very High
Equipment Complexity	Medium	High
Equipment Cost	<\$270	>\$160,000
Skilled Labor Needs	Medium	High
Electricity Costs	Low	High
Sustainability (CO ₂ Production)	None	Very High
Water needs on-site	Low	High

Table 9: Comparison of SandCASTLE vs. other house 3D printers.

My next version of the printer will include a method for automatically depositing the sand for each layer, and improving the resin pump for more reliable operation. Finally, I plan to pursue philanthropic funding to support testing of full-scale prototypes, with the goal of forming a nonprofit for making my house printers available around the world.

1. <https://iconbuild.com>
2. <https://www.apis-corr.com>
3. <https://www.3dwasp.com/en/3d-houses-printing/>
4. <https://www.construction-physics.com/p/3d-printed-buildings>
5. <https://e360.yale.edu/features/can-the-worlds-most-polluting-heavy-industries-decarbonize>
6. <https://homeguide.com/costs/cost-to-frame-a-wall>
7. <https://www.construction-physics.com/p/what-makes-housing-so-expensive>
8. <https://www.construction-physics.com/p/what-progress-has-icon-made-on-3d>